

Soil Microbiome Diversity in Baltic Agricultural Systems

A Comparative Analysis of Conventional and Regenerative Practices

Marta Liepa^{1*}, Jānis Ozols², Ingrid Sørensen³

¹Faculty of Agriculture, University of Latvia, Riga

²Institute of Soil and Plant Sciences, Jelgava

³Department of Agroecology, Aarhus University

marta.liepa@example.org

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Abstract

We compare soil microbiome composition across 24 agricultural sites in Latvia, pairing conventional tillage with regenerative no-till management. Metagenomic sequencing shows significantly higher fungal diversity under regenerative management (Shannon index $H' = 4.2$ versus 2.8, $p < 0.001$), with a marked enrichment of arbuscular mycorrhizal taxa. Nutrient cycling efficiency rose by 35–40% within three growing seasons. The results support regenerative practice as a viable pathway for Baltic agriculture under a cooler, wetter climate.

Keywords: soil microbiome, regenerative agriculture, mycorrhizal networks, Baltic region

1 Introduction

Soil is the most biodiverse habitat on Earth: a single gram can hold billions of microbial cells from tens of thousands of taxa (Bardgett & van der Putten, 2014; Fierer, 2017). These communities regulate decomposition, nutrient availability and soil structure, and they respond quickly to management. Tillage in particular disrupts fungal hyphal networks and accelerates the loss of soil organic carbon (Six et al., 2006, p. 557).

Regenerative agriculture — minimal disturbance, permanent cover and diverse rotations — has been proposed as a remedy, yet most evidence comes from temperate North America (Lal, 2020). Baltic soils, with short growing seasons and high spring moisture, have received little attention.¹ This study asks three questions:

1. Does regenerative management change microbial diversity within three seasons?
2. Which functional groups respond most strongly?
3. Are the shifts associated with measurable gains in nutrient cycling?

2 Materials and methods

2.1 Study sites

We sampled 24 farms across the Zemgale and Vidzeme regions (Table 1). Each regenerative farm was paired with a conventional neighbour on the same soil series, within 5 km, to control for parent material and climate.

Table 1: Study sites by region. SOC is soil organic carbon in the top 15 cm.

Region	Farms	Soil type	Mean SOC (%)	Years under practice
Zemgale	14	Calcaric Cambisol	2.31	3–7
Vidzeme	10	Stagnic Luvisol	2.86	3–5
Total	24		2.54	

2.2 Diversity estimation

Alpha diversity was estimated with the Shannon index,

¹Latvia has roughly 1.9 million hectares of agricultural land, of which about 15% is certified organic.

$$H' = - \sum_{i=1}^S p_i \ln p_i, \quad (1)$$

where S is the number of taxa and p_i the relative abundance of taxon i . Differences between practices were tested with a linear mixed model with farm pair as a random effect:

$$y_{jk} = \beta_0 + \beta_1 x_{jk} + u_j + \varepsilon_{jk}, \quad u_j \sim \mathcal{N}(0, \sigma_u^2) \quad (2)$$

Sequencing followed the protocol of Caporaso et al. (2012), and statistics were computed in R [R Core Team (2025)]:

```
library(lme4)
model <- lmer(shannon ~ practice + (1 | pair), data = samples)
summary(model)
```

3 Results

Regenerative sites showed consistently higher diversity at every sampling location (Figure 1). Using Equation 1, the mean index rose from 2.8 ± 0.2 to 4.2 ± 0.3 ; the mixed model (Equation 2) gave $\beta_1 = 1.38$ (95% CI 1.12–1.64, $p < 0.001$).

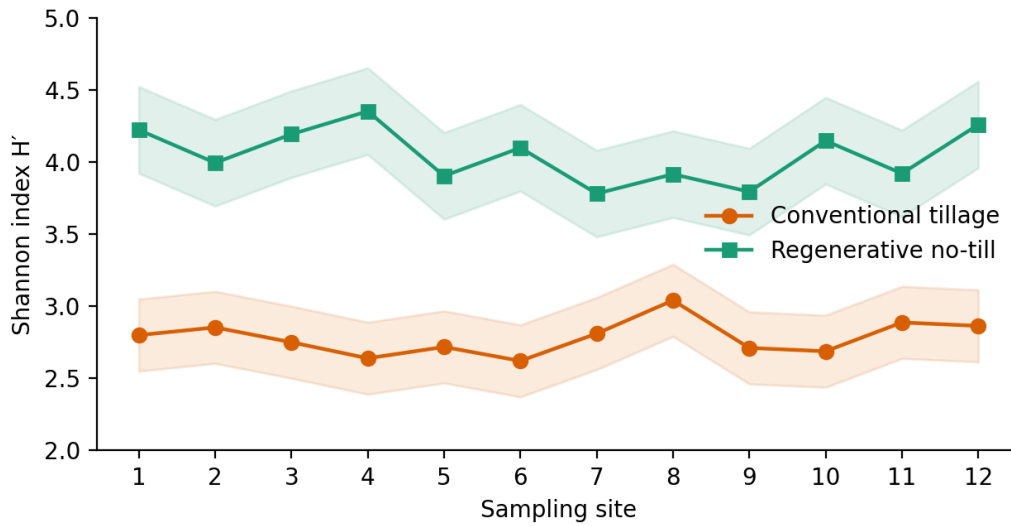


Figure 1: Shannon diversity at the twelve paired sampling sites. Shaded bands show ± 1 standard deviation.

Note

Arbuscular mycorrhizal fungi (Glomeromycota) accounted for most of the difference: their relative abundance was 3.4 times higher under no-till.

4 Discussion

Our findings agree with meta-analyses from warmer climates [Lal (2020) Bardgett and van der Putten (2014)] and extend them to the boreo-nemoral zone. The rapid response – within three seasons – is notable; Six et al. (2006) suggested that aggregate-protected carbon builds over a decade or more.

The soil is the great connector of lives, the source and destination of all. It is the healer and restorer and resurrector.

Two limitations apply. First, sampling covered a single autumn per year. Second, metagenomic read depth limits resolution of rare taxa (see Section 2).

5 Conclusions

Regenerative practice measurably restores soil biological diversity in Baltic conditions within a few seasons. Policy support for the transition — particularly during the first two yield-sensitive years — would accelerate adoption.

References

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